

GLOMERULAR DISEASES

Revision

.Commonly these syndromes are classified in to six category

- .1. Acute nephritic syndrome
- .2. Nephrotic syndrome
- .3. Pulmonary renal syndrome
- .4. Basement membrane syndrome
- .5. Glomerular vascular syndrome

ACUTE NEPHRITIC SYNDROME

- PSGN
- Subacute bacterial endocarditis
- Lupus nephritis
- IGA nephropathy

- Refers to a set of renal disease in which an immunologic mechanism triggers inflammation and proliferation of glomerular tissue that can result in damage to the basement membrane

•ACUTE NEPHRITIC SYNDROME IS CHARACTERIZED BY

- o Hematuria
- o Oliguria
- o fluid retention manifested by –edema and hypertension
- o ARF, manifested by sudden decrease in the GFR

POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

• The most common cause of GN in children. • Group A β -hemolytic streptococcal infections are common in children and can lead to the postinfectious complication of Acute Glomerulonephritis (GN).

•Acute Poststreptococcal Glomerulonephritis (APSGN)
is a classic example of the

•acute nephritic syndrome characterized by:

•o The sudden onset of gross hematuria,

•o Edema,

•o Hypertension, and

•o Renal dysfunction.

•It is one of the most common glomerular causes of gross hematuria in children.

•• APSGN follows infection of the throat or skin by certain nephritogenic strains of group A β -hemolytic streptococci.

CLINICAL MANIFESTATIONS

- Poststreptococcal GN is most common in children ages 5-12 yrs.
- The typical patient develops an acute nephritic syndrome 1-2 wk. after an antecedent streptococcal pharyngitis or 3-6 wk. after a streptococcal pyoderma.

INVESTIGATIONS

- U/A
- o Hematuria
- o Red blood cell cast
- o Proteinuria in nephritic range
- o Polymorphonuclear leukocytes

- CBC

- Anemia (usually hemodilution and low-grade hemolysis)

- Confirmation of the diagnosis requires clear evidence of a prior streptococcal infection
- A positive throat culture report might support the diagnosis or might
- represent the carrier state. A rising antibody titer to streptococcal antigen(s) confirms a recent streptococcal infection.
- ▪ The anti-streptolysin O titer is commonly elevated

COMPLICATIONS

- Acute complications result from hypertension and acute renal dysfunction.
- Electrolyte abnormality
 - Hyperkalemia,
 - Hyperphosphatemia,
 - Hypocalcemia,
 - Acidosis,

TREATMENT

- Generally, the management is supportive.
- 10-day course of systemic antibiotic therapy with Penicillin
- Diuretics-frusemide
- Salt restriction and fluid restriction

PROGNOSIS

- Complete recovery occurs in $> 95\%$ of children with APSGN.

NEPHROTIC SYNDROME

- Nephrotic syndrome is the clinical manifestation of glomerular diseases associated with heavy (nephrotic-range).

• Characterized by:

- Hypoalbuminemia (≤ 2.5 g/dL),
- Heavy (nephrotic-range) proteinuria
- Edema, and
- Hyperlipidemia (cholesterol > 200 mg/dL)

Cause

- Minimal change disease
- Focal segmental
glomerulosclerosis
- Membranous nephropathy

- Most common is minimal change disease about 85%

CLINICAL CONSEQUENCES OF NEPHROTIC SYNDROME

.EDEMA

.HYPERLIPIDEMIA

- an increase in Cholesterol,
- o Increase in Triglycerides,
- Increase in Low-Density Lipoproteins (LDL),
and
- Very-Low-Density Lipoproteins (VLDL

- o Hypoglobulinemia
- (mainly) as a result of the urinary losses of immunoglobulin IgG.
- HYPERCOAGULABILITY

DIAGNOSIS

- The diagnosis of nephrotic syndrome is confirmed by urinalysis with the first morning urine protein
- The urinalysis reveals 3+ or 4+ proteinuria, and microscopic hematuria is present in 20% of children

- The serum albumin level is < 2.5 g/dL,
- serum cholesterol and triglyceride levels are elevated

Treatment

.USE OF CORTICOSTEROIDS

- .Prednisone or prednisolone should be administered as a single daily dose of 60 mg/m² /day or 2 mg/kg/day to 60 mg daily (max) for 4-6 wk Followed by alternate-day prednisone (starting at 40 mg/m² qid or 1.5 mg/kg qid) for a period ranging from 8 wk to 5 mo, with tapering of the dose.

.80–90% of children respond to steroid therapy.

How you difference

- Proteinuria
- Hyperlipidemia and lipiduria with faulty casts
- Hypertension volume of urine passed in 24hr
- Hematuria

•AKI

•Relapse

•Read more,,,,,,,,,,,,,